

Total No. of Questions : 8]

PD4208

SEAT No. :

[Total No. of Pages : 2

[6403]-1

T.E. (Civil Engineering)

**HYDROLOGY AND WATER RESOURCES ENGINEERING
(2019 Pattern) (Semester - V) (301001)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answers Q.No 1 or Q.No 2, Q.No 3 or Q.No 4, Q.No 5 or Q.No 6, Q.No 7 or Q.No 8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1) a)** State the various methods to estimate the magnitude of peak flood. Explain any two methods in detail. **[2+7]**
- b)** Estimate the maximum flood flow for the following catchments by using an appropriate empirical formula. **[9]**
Consider Dickens constant 6.0 and Ryves Coefficient 6.8.
- i) $A_1 = 40.5 \text{ km}^2$, (for Western Ghat area, Maharashtra).
 - ii) $A_2 = 40.5 \text{ km}^2$, (in Gangetic plain).
 - iii) $A_3 = 40.5 \text{ km}^2$, (in the Cauvery delta, Tamil Nadu).

OR

- Q2) a)** State and explain Q-GIS software application in hydrology (watershed delineation). **[8]**
- b)** What is hydrologic flood routing? Explain Muskingum method in detail. **[3+7]**

- Q3) a)** Explain in brief various investigations required for reservoir planning. **[7]**
- b)** State the significance of mass curve. Explain the step by step procedure for fixation of reservoir capacity using annual inflow and outflow. **[2+8]**

OR

- Q4) a)** Explain with neat sketch the significance of reservoir sedimentation and trap efficiency. **[5+5]**
- b)** What are the different losses which take place from the reservoir. Explain the measures to control this losses. **[3+4]**

P.T.O.

Q5) a) Derive the formula to calculate discharge of a well in a confined aquifer and unconfined aquifer. **[10]**

b) Write the preventive and curative measures of water logging. Explain tile drain method and also state the formula for spacing of tile drains. **[3+5]**

OR

Q6) a) A 30 cm well completely penetrates an unconfined aquifer of saturated depth 40m. After a long period of pumping at steady rate of 1500 lpm, the drawdown in two observation wells 25 and 75m from the pumping well were found to be 3.5 and 2 m respectively. **[5+5]**

i) Determine the transmissivity of the aquifer.

ii) What is the drawdown at pumping well?

b) What is Participatory Irrigation Management (PIM)? What is co-operative water distribution systems. **[4+4]**

Q7) a) Write the advantages and disadvantages of PDN over conventional canal distribution network. **[10]**

b) Differentiate between surface irrigation and subsurface irrigation and explain sprinkler irrigation in detail. **[7]**

OR

Q8) a) State various methods of canal revenue collection and explain any two in detail. **[10]**

b) What is micro irrigation and what are its advantages compared other methods of irrigation. **[3+4]**

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